

EXP-008 — Chunk Size × Dense Retrieval

Testing whether the chunk-length correlation found in EXP-007 was causal, or only a proxy.

EXECUTIVE RESULT

The interaction hypothesis was not supported, and the EXP-007 chunk-length correlation was not causal. Bounded chunking made dense retrieval slightly worse (0.425 → 0.400), collapsed MRR from 0.360 to **0.259**, and left deep reachability unchanged at 5 spans absent@300. Paired: 1 rescued, 1 regressed, **net 0** — and both are single-rank boundary crossings.

The interaction runs the *wrong way*: chunk size moved BM25 **+0.025** and dense **−0.025**, an interaction of **−0.050**.

−0.050

Interaction term. The hypothesis predicted a positive interaction; the measurement is negative

AN-003: 119

The one genuine gain — the answer chunk entered the dense pool for the first time, but at rank 119, offset by AN-004 being lost entirely

Intervention fidelity

The C → D comparison is an isolation of chunk size only if nothing else moved. Gated and verified: same 202 document versions, chunk bodies are exact source substrings in both, **zero** enriched rows or V3 transforms in either, and **22/22** evidence spans map in both. Critically the **token vocabulary is identical** — 32,011 forms, zero bounded-only — because V2 is a re-partition of the same text with boundaries on whitespace. The embedding model presents exactly the same tokens; only the boundaries differ.

chunk set	chunks	mean	median	p90	max	>2,000
control	14,209	1,097	508	3,466	16,096	3,069
bounded	20,526	757	917	1,193	1,999	0

Bounded embedding build used the same model and fingerprint: 20,526 vectors, token match rate 0.9627 (identical to control), all-zero embeddings 117 → 128 — a *fall* in rate from 0.82% to 0.62% across 44% more chunks.

The 2×2

Cell	macro	full	spans@10	doc	MRR	a@10	a@20	a@50	a@100	a@300	ms
A — control chunks + BM25	0.475	9/20	10/22	0.825	0.280	12	8	3	2	1	384
B — bounded chunks + BM25	0.500	9/20	11/22	0.825	0.287	11	9	5	4	1	395

Cell	macro	full	spans@10	doc	MRR	a@10	a@20	a@50	a@100	a@300	ms
C — control chunks + dense	0.425	8/20	9/22	0.725	0.360	13	9	9	7	5	90
D — bounded chunks + dense	0.400	8/20	8/22	0.675	0.259	14	11	10	10	5	82

All three reproduction gates pass: **A** reproduces the frozen BM25 baseline, **B** reproduces EXP-005A, **C** reproduces EXP-007B. Bounded chunking made dense worse at every depth except 300, where it is unchanged — there is no depth at which the intervention improved candidate recall.

Interaction

Retriever	control	bounded	Δ	fully recalled	absent@300	MRR	net
BM25	0.475	0.500	+0.025	9 → 9	1 → 1	0.280 → 0.287	+0
dense	0.425	0.400	-0.025	8 → 8	5 → 5	0.360 → 0.259	+0

Neither retriever gained a single fully-recalled case, and both kept their absent@300 count exactly.

Paired C → D span movements

Movement	n
worsened no cross	6
unchanged	6
still unreachable	4
newly reachable outside k	1
lost entirely	1
strong regression	1
improved no cross	1
boundary improvement	1
boundary regression	1

Six spans worsened without crossing the cutoff against one that improved. That is the signature behind the MRR collapse. Mean/median evidence rank when found: **C 34.1 / 10, D 56.5 / 11**.

AN-003 — the canonical case

Cell	rank	doc rank	chunk len	@10	@20	@50	@100	@300
A — control chunks + BM25	—	6	—	no	no	no	no	no
B — bounded chunks + BM25	—	4	—	no	no	no	no	no
C — control chunks + dense	—	2	—	no	no	no	no	no
D — bounded chunks + dense	119	3	1191	no	no	no	no	yes

The anchor chunk fell **3,449** → **1,191** characters and, for the first time in any experiment, the answer-bearing chunk became retrievable. **This is partial support, and only barely:** rank 119 is an order of magnitude outside `top_k` and outside any pool a reranker would consume. It changed no recall at any depth ≤ 100 , and it was paid for exactly — AN-004 went from rank 17 to **absent@300**, so the corpus-wide count stayed at 5.

What shorter chunks did to dense's wins

Case	A bm25/ctrl	B bm25/bnd	C dense/ctrl	D dense/bnd
AN-002	27	172	1	6
AN-007	18	34	2	6
AN-012	47	6	1	140
AN-004	12	11	17	—

EXP-007's two dense rescues both lived in the same 3,327-character `HTTP errors` chunk — a topically homogeneous list of status codes. Both degrade (1 → 6, 2 → 6), and AN-012 collapses from **1 to 140**.

MECHANISM

Mean pooling rewards **topical coherence**, not shortness. A long chunk uniformly about one topic produces a strong clean vector; splitting it fragments that signal across weaker ones. Chunk length in EXP-007 was a *proxy* for heterogeneity, not the cause — long chunks here are sometimes coherent (`HTTP errors`) and sometimes not (`Body Parameters`). Uniform shortening helps the heterogeneous minority slightly and damages the coherent majority.

The correlation survives bounded chunking, which confirms shortening cannot fix it: in D, median chunk length is **883** for reachable evidence and **1,167** for unreachable — still $\sim 1.3\times$, on a corpus where nothing exceeds 1,999.

Exploratory fusion — EXP-008E

The precondition ("if bounded dense improves") was **not met**, so this is exploratory only. Preregistered EXP-007 settings, untuned.

Configuration	macro	full	doc	MRR	absent@300
A — BM25 control alone	0.475	9/20	0.825	0.280	1

Configuration	macro	full	doc	MRR	absent@300
E — BM25 control + bounded dense (RRF)	0.475	9/20	0.775	0.278	2
<i>EXP-007C — BM25 control + control dense</i>	<i>0.600</i>	<i>11/20</i>	<i>0.825</i>	<i>0.326</i>	<i>2</i>

Fusion collapsed to baseline. EXP-007's fusion reached 0.600 because control dense contributed complementary rankings; degrading dense destroyed that complementarity — strong indirect confirmation that bounded chunking hurt the dense retriever. The one positive: the prior fusion regression improved, OA-004 moving from RRF rank 17 to **9**, back inside `top_k`.

Project state — four hypotheses, four negative results

#	Hypothesis	Verdict
1	Oversized chunks hide evidence (BM25)	falsified — 0 rescued
2	Missing structural context	falsified — $\Delta 0.000$
3	Lexical vocabulary mismatch	unsupported — no vocabulary rescue
4	Chunk size interacts with dense retrieval	not supported — interaction -0.050

The best measured configuration remains **EXP-007C** (BM25 + dense, both on control chunks, 0.600 / 11-of-20). Both retrievers work better on the original chunking than on the bounded one.

Limitations

1. **n = 20**. One case is five percentage points. Every movement is 1–2 cases; no significance is claimed.
2. **One dense model, and a weak one**. Static word vectors with mean pooling, because transformer hosts remain egress-blocked. Mean pooling is very likely why coherence matters this much. **This constrains mean-pooled static embeddings, not dense retrieval in general.**
3. **One bounded configuration** (target 1,200 / cap 2,000, chosen in EXP-005 from corpus percentiles). No sweep was run, deliberately.
4. **EXP-008E is exploratory** and its precondition was not met.
5. **EXP-NUL** still has not run.

What is justified next

1. **Stop optimising chunk size**. Two controlled interventions on two retrievers, both null.
2. **A transformer retrieval encoder** remains the highest-value experiment, blocked only on egress. EXP-008 sharpens why: mean pooling is now directly implicated, and a transformer is exactly the intervention that removes it.
3. **If chunking is ever revisited, test topical homogeneity, not length** — but behind (2).

4. **Still not a reranker.** AN-003 sits at rank 119 in the one cell where it is reachable and absent@300 in the other three.

Promotion decision: the frozen baseline does not change (control chunking, no enrichment, BM25, top_k=10). Bounded chunking is not promoted for dense retrieval; it is measurably worse. The EXP-007 hybrid remains the leading candidate, and EXP-008 shows it must be built on **control** chunks for both retrievers.

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